

Advanced Topics in Monetary Economics with Heterogenous Agents

— Problem Set 2, Spring 2021 —

Due: Monday, Feb 15, 8:29 AM ET (before Section) on Canvas

Important guidelines.

1. Please only submit this problem set if you are *not auditing* this course and are *officially enrolled* for credit. Each problem set counts for 20% of your final grade.
2. Make sure to *submit your code*. The best way to solve our problem sets is using Jupyter Notebooks. We *strongly suggest using Python* from the beginning since this course will progressively build on existing Python modules in future problem sets. You can ask your classmates coding questions in Slack, Martin in OH, and review QuantEcon's programming tutorial.¹
3. The problem sets for this class are meant to give you exposure to the actual, messy research process. This means: If a question is incomplete or missing information, please go ahead and make an assumption that you deem reasonable. Be sure to *state your assumptions and submit your full derivations* in your solution. Many problems could have multiple solutions.

1 The Intertemporal Keynesian Cross with OLG Agents

Consider the following OLG model. Agents live for three periods. They are first young y , then middle-aged m , then old o . Each generation has mass 1. Middle-aged agents earn an average income of Z_t , corresponding to aggregate after-tax income in period t .

An agent solves

$$\max (1 - \beta) \log c_{t-1}^y + \beta(1 - \beta) \log c_t^m + \beta^2 \log c_{t+1}^o$$

subject to a standard present value budget constraint

$$c_{t-1}^y + \frac{c_t^m}{1+r} + \frac{c_{t+1}^o}{(1+r)^2} \leq \frac{Z_t}{1+r}$$

r is the constant real interest rate in the economy. Call Z the steady state value of Z_t . Middle-aged agents at date 0 solve

$$\max (1 - \beta) \log c_t^m + \beta \log c_{t+1}^o$$

¹<https://python-programming.quantecon.org/intro.html>

$$c_t^m + \frac{c_{t+1}^o}{1+r} \leq Z_0$$

And old agents at date 0 follow their steady state consumption.

The rest of the economy is exactly as in Lecture 3, only the household side is different.

1. Write down the optimal consumption policies of a generation born at some date $t - 1 \geq 0$, as well as the optimal consumption policies of agents that are middle aged at date 0 and that are old at date 0.
2. Write down the aggregate consumption function $C_t = C_t(\{Z_s\})$ for each period $t \geq 0$.
3. Find the matrix of intertemporal MPCs $\mathbf{M}$. Verify that its columns add to 1 in present value terms. Note that $\mathbf{M}$ is *asymptotically Toeplitz*, that is for large s we have that

$$\sup_m |M_{s+k,s} - M_{s+k+m,s+m}| \rightarrow 0$$

for any k .

4. Define $\eta \equiv \beta^2/(1 - \beta)$. Compute a right-eigenvector of $\mathbf{M}$ with eigenvalue 1. (Hint: η should feature in it.) Call that eigenvector $\mathbf{v}$.

Next we solve the intertemporal Keynesian cross (IKC)

$$d\mathbf{Y} = d\mathbf{G} - \mathbf{M}d\mathbf{T} + \mathbf{M}d\mathbf{Y}$$

for $d\mathbf{Y}$. We choose a specific path for fiscal policy, namely

$$d\mathbf{G} = (1, 0, \dots)'$$

$$d\mathbf{T} = (0, 1 + r, 0, \dots)'$$

By design, $d\mathbf{G} - d\mathbf{T}$ has zero NPV.

5. Guess and verify that one solution to the IKC is given by [NOTE: the original statement seemed to have a typo]

$$d\mathbf{Y}^* = (\beta^{-1}, (1 + r), 0, 0 \dots)$$

Show that $d\mathbf{Y}^*$ has positive NPV. Show that for any scalar $d\lambda$, $d\mathbf{Y}^* + d\lambda \cdot \mathbf{v}$ is also a solution to the IKC.

6. Show that $d\mathbf{Y}^*$ is *locally determinate* in that it is in fact the only bounded solution to the IKC if and only if $\eta(1 + r) > 1$.

We next ask: How come $d\mathbf{Y}^*$ has **positive** NPV while the “partial equilibrium” demand shock we feed into the economy, $d\mathbf{G} - \mathbf{M}d\mathbf{T}$, has zero NPV. To do so, recall that we saw in

class that we can decompose $d\mathbf{Y}^*$ into

$$d\mathbf{Y}^* = \sum_{k=0}^{\infty} \mathbf{M}^k (d\mathbf{G} - \mathbf{M}d\mathbf{T}) + d\lambda \cdot \mathbf{v}$$

for a specific $d\lambda$. Define

$$d\mathbf{Y}^{(K)} \equiv \sum_{k=0}^K \mathbf{M}^k (d\mathbf{G} - \mathbf{M}d\mathbf{T})$$

$$d\mathbf{Y}^{(\infty)} \equiv \lim_{K \rightarrow \infty} \sum_{k=0}^K \mathbf{M}^k (d\mathbf{G} - \mathbf{M}d\mathbf{T})$$

corresponding to the path of output after K rounds of the Keynesian cross.

- i. Show that $d\mathbf{Y}^{(K)}$ is always zero NPV, for any K .
- ii. Show that, if $\eta = \beta = 0$, $d\mathbf{Y}^{(\infty)}$ is well defined and has zero NPV. Derive $d\mathbf{Y}^{(\infty)}$. Compute $d\lambda$ that is needed so that $d\mathbf{Y}^* = d\mathbf{Y}^{(\infty)} + d\lambda \cdot \mathbf{v}$.
- iii. Show that, if $\eta = \infty$ and $\beta \nearrow 1$, $d\mathbf{Y}^{(\infty)}$ is well defined and has positive NPV. Derive $d\mathbf{Y}^{(\infty)}$.
- iv. More generally, illustrate numerically using specific examples (choose your own parameters) that whenever $\eta < 1$, $d\mathbf{Y}^{(\infty)}$ seems to have zero NPV, and whenever $\eta > 1$, $d\mathbf{Y}^{(\infty)}$ seems to have positive NPV.

[Feel free to submit a theoretical proof of this. But it's a very hard problem. As a hint, the theory of Markov chains and "charges" on Markov chains helps. Happy to give you references if you're curious.]

References: a math book called "Denumerable Markov Chains" by Kemeny Snell & Knapp.

In summary we have seen that whenever

- $\eta < 1$, the positive NPV nature of $d\mathbf{Y}^*$ is coming from $d\lambda \cdot \mathbf{v}$.
- $\eta > 1$, the positive NPV nature of $d\mathbf{Y}^*$ emerges endogenously from $d\mathbf{Y}^{(\infty)}$, despite all $d\mathbf{Y}^{(K)}$ having zero NPV!

What is η more generally, in an economy with a richer $\mathbf{M}$? To hint at this, consider the following exercise.

7. Consider a far-out column of $\mathbf{M}$ (i.e. $M_{\cdot, s}$ for large s). Show that

$$\sum_{t=0}^{\infty} (1+r)^{s-t} M_{t,s} (t-s) > 0$$

if and only if $\eta > 1$. *[This is the general condition that holds in a richer $\mathbf{M}$ that is still asymptotically Toeplitz.]*

2 A Model of Firm Dynamics

2.1 Part I: No entry and exit

This question asks you to simulate a version of the Hopenhayn-Rogerson model.

- There is a representative household

$$\max \sum_{t=0}^{\infty} \beta^t \log C_t$$

$$C_t + A_t \leq (1 + r_{t-1})A_{t-1} + w_t N$$

where $N = 1$ is an exogenous labor endowment.

- There are heterogeneous firms. A firm is labeled by idiosyncratic productivity s following a discretized AR(1) process and current employment n_- . Firms solve the Bellman equation

$$\max V_t(s, n_-) = \max_n \left\{ d + \frac{1}{1 + r_t} \mathbb{E} [V_{t+1}(s', n) | s] \right\} \quad (1)$$

- Initially, we assume firms neither die nor enter, they simply exist and hire labor to make flow profits

$$d = f(s, n_-) - w_t n_- - \Psi(n, n_-)$$

- where the production function depends on α, X_t .

$$f(s, n_-) = X_t s^\alpha n_-^{1-\alpha}$$

- and where $\psi > 0$ gives the strength of labor adjustment costs

$$\Psi(n, n_-) = \frac{\psi}{2} \left(\frac{n}{n_-} - 1 \right)^2 n_-$$

- Their distribution is denoted $D_t(s, n_-)$ and their employment policy is given by $n_t^*(s, n_-)$. Aggregate labor demand by firms is then $N_t^f = \int n_- dD_t(s, n_-)$. Aggregate production is $Y_t = \int f(s, n_-) dD_t(s, n_-)$ and aggregate adjustment costs are $\Psi_t = \int \Psi(n^*(s, n_-), n_-) dD_t(s, n_-)$. Aggregate dividends are $\Phi_t = \int d(s, n_-) dD_t(s, n_-) = Y_t - w_t N_t^f - \Psi_t$. Let A_t^f be the *total value of firms*, which you will work out below.

For all numerical exercises, use the following calibration:

- $\beta = 0.8$. $\alpha = 0.66$. $Y = 1$. $\psi = 4$. $\log s$ follows an AR(1) with persistence $\rho = 0.9$ and standard deviation of the shocks of 0.2. Discretize it using Rouwenhorst methods over at least

7 points. Make sure to define n_- on a grid of at least 100 points. Choose the minimum and maximum points on the n_- grid judiciously (see section notes if needed).

1. What is the equilibrium interest rate r_t paid to savers for holding bonds in this economy, and what is its value in the steady state?
2. Since in addition to bonds households can also put their wealth in firm stocks, write down a *no arbitrage* condition defining equilibrium A_t^f , the total value of firms in the economy. [Hint: you can approach this question deriving two Euler equations for the representative household, or simply reasoning with economic intuition].
3. In equilibrium, we have that (1) the household optimizes, (2) firms optimize, and (3) markets clear:

$$A_t = A_t^f \quad (2)$$

$$N = N_t^f \quad (3)$$

$$C_t = Y_t - \Psi_t \quad (4)$$

Prove that the goods market clearing condition (4) follows the other equilibrium conditions, including asset and labor market clearing (2) and (3).

4. Write down the first order condition of the Bellman equation (1).
5. Write down the Envelope condition of the Bellman equation (1).
6. Use those two conditions together with the endogenous grid point method for a backward iteration to find the steady state employment policy function $n^*(s, n_-)$. Assume some values of w, Z . Plot the policy function for various s .
7. Compute the steady state distribution using a forward iteration. Plot it for various s .
8. Compute aggregates N^f, Y, Ψ, Φ , as well as the average value function $\int V(s, n_-) dD(s, n_-)$. [Hint: simply carry it in your backward iteration]
9. Plot w as a function of $N^f(w)$, incumbent labor demand.
10. Now we calibrate the model: Find TFP X and wage w so that $Y = 1$ and $N^f = 1$.

2.2 Part II: Exogenous entry and exit

Now we modify the problem to allow for entry and exit. Entrants take one period to come online and be able to produce, much like capital investment. We assume that there is a probability $\delta > 0$ with which a firm exits. Moreover, there is an exogenous (but possibly time-varying) entry rate $m_t > 0$. The Bellman equation is now given by

$$\max V_t(s, n_-) = \max_n \left\{ f(s, n_-) - w_t n_- - \frac{\psi}{2} \left(\frac{n}{n_-} - 1 \right)^2 n_- + \frac{1}{1 + r_t} \mathbb{E}_{s'} [V_{t+1}(s', n) | s, \text{not exit}] \right\}$$

Upon entry, assume all entrants are injected in a single point far down at the bottom of the employment grid, i.e. $G(s, n_0)$. However, the first s draw is taken from the stationary distribution of the Markov process.

1. Write down a Bellman equation where δ shows up, and modify the backward iteration to allow for $\delta > 0$.
2. Modify the forward iteration to allow for m_t .
3. Solve for the steady state with entry and exit, assuming $\delta = 0.1$, with entry rates being 20% higher than exit rates. [Note: Check that the mass of firms in the economy coincides with what you expected].
4. Compute aggregates. Compute the *expected value* from entering, $V = \int V(s, n_-) dG(s, n_-)$.

2.3 Part III: Endogenous entry

Now we allow for endogenous entry. Assume that entry requires labor $n^e = 1$ per each firm m that enters paid in labor.

1. Write down the new labor market clearing equation and the free entry condition. [Hint: we assume that entrants take one period to come online and be able to produce, much like capital investment. Make sure free entry reflects this fact].
2. Calibrate the new steady state with free entry: Find TFP X , wage w , and entry rate m such that $Y = 1$, $N = 1$, and the two conditions you wrote above hold.
3. Reduce N by 10%. Solve for the new steady state wage w and entry rate m such that the two conditions you write down above still hold. Interpret the result in light of recent reductions in population growth.

2.4 Part IV: Stochastic Simulation

To do this, draw a sample of 5000 firms who resemble the average entrant in the economy. To perform a stochastic simulation, initiate each of those firms at the same point, the state of the average entrant, and simulate $T = 50$ forward in time. Make sure in each period you draw from an *i.i.d.* death probability δ ; if the firm dies, it stops producing. If it survives, they obtain a new shock s' given their current s ; you should draw using the right row from the matrix Π each time. Then, using the optimal policies you derived above, track the evolution of the endogenous state n_- as the firm lifecycle progresses.

Plot the evolution of average employment, TFP, and sales for the average firm that enters the economy. Plot the share of survivors in your simulation, as well as the employment history of each of the firms in your sample. Feel free to provide any other moments on the firms in your simulation that you deem interesting.